BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Utility Software

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What does operating-system file management do?

2. Why split network protocols into layers?
   A) All layers must be identical
   B) No interfaces are needed
   C) Every packet becomes encrypted
   D) Each layer has a job

3. "Encryption alone guarantees files cannot be stolen or deleted." Correct this. Explain why.

4. A fake useful app locks files for a ransom. "Malware is any app that uses lots of RAM." Correct this. How does this malware cause harm?

Today's objectives

Explain the core ideas of utility software using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Utility software

Carries out system maintenance or security tasks alongside the operating system.

Encryption utility

Protects stored data by transforming it into ciphertext.

Compression utility

Reduces file sizes for storage or transmission.

Defragmentation

Reorganises file fragments on spinning magnetic disks; not normally beneficial for SSDs.

BLACK PEN: Write one definition in your own words.

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Learn and practise

New Learning - Key ideas

Utility software
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Defragmentation
Reorganises file fragments on spinning magnetic disks; not normally beneficial for SSDs.

I Do - Worked model

A school has a slow magnetic HDD with heavily fragmented files. Explain defragmentation.

- Explain file fragments may be spread across disk positions.
- Defragmentation rearranges related blocks contiguously.
- This can reduce mechanical seek movement.
- Avoid recommending it as a routine performance fix for SSDs.

We Do - Guided practice

1. A teacher must email a folder of large documents. Select a utility.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Write the calculation for saving storage when a file is compressed.

```
originalMB = 500
compressedMB = 350
saved = ...
print(saved)
```

You Do - Independent exam questions

Q2. State a purpose of encryption utility software. [2 marks]

Q3. Explain why defragmentation can improve HDD access. [3 marks]

Q4. Explain why lossless compression matters for source code. [3 marks]

Exit Ticket - True or False?

Defragmenting an SSD routinely improves it in the same way as an HDD.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. It manages files and folders: naming, saving, copying, moving and deleting them.

2. D - Each layer has a job

3. Without the key, an attacker cannot easily read ciphertext. Encryption does not stop theft or deletion.

4. Malware is made to harm, disrupt or gain access without permission. Different types work in different ways and can affect data access, privacy or availability.

I Do - Explain the model

Explain file fragments may be spread across disk positions.

Defragmentation rearranges related blocks contiguously.

This can reduce mechanical seek movement.

Avoid recommending it as a routine performance fix for SSDs.

We Do - Model answers

1. Note lossless methods when exact files are required.

Code First - Answer

```
originalMB = 500
compressedMB = 350
saved = originalMB - compressedMB
print(saved)
```

Expected output: 150

You Do - Self-mark

Q2: Converts readable data into ciphertext to restrict unauthorised reading.

Q3: It places related fragments closer together, reducing mechanical seek time.

Q4: Original bytes must be recovered exactly; lossy removal could corrupt instructions.

GREEN PEN: Choose the hardest answer. Correct it on lined paper.